Econometrics I
PPS-03
Multiple Choice Questions
(chapters 4-5-6)
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1. Asymptotic ..... means that in large samples OLS estimator follows ..... distribution. This implies that we do not need the ..... assumption for the errors. **Fill in the blanks.**

   A. normality / normal / consistency
   B. consistency / normality / normal
   C. consistency / consistent / consistency
   D. normality / normal / normality
   E. normality / exponential / consistency

2. Unrestricted model is given by

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7 + u$$

   Restricted model is

$$y = \beta_0 + \beta_1 x_1 + \beta_3 x_3 + \beta_5 x_5 + \beta_7 x_7 + u$$

   Find the null hypothesis.

   A. $H_0 : \beta_1 = 0,\ \beta_3 = 0,\ \beta_6 = 0$
   B. $H_0 : \beta_1 = 0,\ \beta_3 = 0,\ \beta_7 = 0$
   C. $H_0 : \beta_2 = 0,\ \beta_4 = 0,\ \beta_6 = 0$
   D. $H_0 : \beta_2 = 0,\ \beta_4 = 1,\ \beta_6 = -1$
   E. $H_0 : \beta_2 = -1,\ \beta_4 = 0,\ \beta_6 = 1$

3. **If** $SSR_{UR} = 520$**,** $SSR_R = 640$ **, number of restrictions=**3 **and** $n - k - 1 = 260$**, F statistic equals ..... .**

   A. -2
   B. 2
   C. -20
   D. 20
   E. -0.02

4. Consider the following regression model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + u$$

   You want to test the null hypothesis $H_0 : \beta_1 = 2\beta_3, \beta_4 = -1, \beta_6 = 0$.

   What is the restricted model under the null?

   A.
$$y + x_4 = \beta_0 + \beta_2 x_2 + \beta_3 (2x_1 + x_3) + \beta_5 x_5 + u$$

   B.
$$y + x_4 = \beta_0 + \beta_1 (x_1 + 2x_3) + \beta_2 x_2 + \beta_5 x_5 + u$$

   C.
$$y - x_4 = \beta_0 + \beta_1 (2x_1 + x_3) + \beta_2 x_2 + \beta_5 x_5 + u$$

   D.
$$y - x_4 = \beta_0 + \beta_1 (x_1 + 2x_3) + \beta_2 x_2 + \beta_5 x_5 + u$$

   E.
$$y - x_4 = \beta_0 + \beta_2 x_2 + \beta_3 (x_1 + 2x_3) + \beta_5 x_5 + u$$

5. The following is the R regression output for the PRF:

$$educ = \beta_0 + \beta_1 fatheduc + \beta_2 motheduc + \beta_3 sibs + \beta_4 abil + u$$

Variable definitions: `educ` is the education level measured in years, `fatheduc` and `motheduc` are the education levels of father's and mother's, respectively, both measured in years, `sibs` is the number of siblings, and `abil` is a measure of cognitive ability.

```
lm(formula = educ ~ fatheduc + motheduc + sibs + abil)
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  8.97212    0.31766  28.244  < 2e-16
fatheduc     0.150      0.030    ...... 2.04e-08
motheduc     0.173      0.029     5.966 1.86e-09
sibs        -0.109      0.028    -3.893 0.000102
abil         0.500      0.025    20.000  < 2e-16
---

Residual standard error: 1.773 on 1220 degrees of freedom
Multiple R-squared:  0.50
F-statistic: .............. on ......... and ........... DF
```

What is the F statistic for the overall significance of the regression model? What are the degrees of freedom?

A. $F = 305$ with 5 numerator degrees of freedom and 1221 denominator degrees of freedom.

B. $F = 305$ with 4 numerator degrees of freedom and 1219 denominator degrees of freedom.

C. $F = 305$ with 4 numerator degrees of freedom and 1220 denominator degrees of freedom.

D. $F = 303.75$ with 4 numerator degrees of freedom and 1220 denominator degrees of freedom.

E. $F = 303.75$ with 5 numerator degrees of freedom and 1220 denominator degrees of freedom.

6. The following is the R regression output for the PRF:

$$educ = \beta_0 + \beta_1 fatheduc + \beta_2 motheduc + \beta_3 sibs + \beta_4 abil + u$$

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---

Residual standard error: 1.773 on 1220 degrees of freedom
Multiple R-squared:  0.50
```

A researcher is interested in testing $H_0 : \beta_1 = 0, \beta_2 = 0$, that is, if `motheduc` and `fatheduc` are jointly significant or not, using the appropriate test statistic. The R output for the restricted model under the null hypothesis is given as follows:

```
lm(formula = educ ~ sibs + abil)
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 12.38650    0.11618  106.62  < 2e-16
sibs        -0.15572    0.02910   -5.35 1.05e-07
abil         0.61788    0.02482   24.89  < 2e-16
---

Residual standard error: 1.874 on 1222 degrees of freedom
Multiple R-squared:  0.40
```

Compute the F statistic for $H_0 : \beta_1 = 0, \beta_2 = 0$ and state its distribution with the appropriate degrees of freedom.

A. $F = 122 \ \sim \ F(2, \ 1222)$

B. $F = 203.67 \ \sim \ F(3, \ 1220)$

C. $F = -122 \ \sim \ F(2, \ 1220)$

D. $F = 122 \ \sim \ F(2, \ 1220)$

E. $F = 611 \ \sim \ F(2, \ 1222)$

7. **Which of the following is used to exploit decreasing or increasing ceteris paribus effect of a explanatory variable on the independent variable in a regression?**

A. Models with only one dummy variable

B. Models with variables in level

C. Models with several dummy variables

D. Models with quadratic functions

E. Models with interaction terms

8. **Fill in the blank: If we change the unit of measurement of the dependent variable then ..... will not change.**

A. SSR B. SST C. t statistics D. SSE E. Coefficient estimates

9. **Consider the following model:**

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 \times x_2 + u$$

where $x_1 \times x_2$ is an interaction term. Find an expression for the marginal impact of x_1 on y, assuming $\Delta u = 0$.

A.
$$\frac{\Delta y}{\Delta x_1} = \beta_1$$

B.
$$\frac{\Delta y}{\Delta x_1} = \beta_1 + \beta_2 x_2$$

C.
$$\frac{\Delta y}{\Delta x_1} = \beta_1 + \beta_3 x_2$$

D.
$$\frac{\Delta y}{\Delta x_1} = \beta_1 + \beta_3 x_1$$

E.
$$\frac{\Delta y}{\Delta x_1} = \beta_2 + \beta_3 x_1$$

10. **Consider the following logarithmic wage equation**

$$\widehat{\log(\text{wage})} = 6.01 + 0.04 \text{ educ} + 0.004 \text{ educ} \times \text{pareduc}$$

$$n = 684, \qquad R^2 = 0.21,$$

where educ is the number of schooling year, and pareduc is the sum of education of parents (years). For an individual with pareduc=20, what is the predicted return to education?

A. 6.01%

B. 12%

C. 4%

D. 4.4%

E. 84%

11. **Consider the following models:**

$$\text{Model A}: \quad y = \beta_0 + \beta_1 x + u$$

$$\text{Model B}: \quad y = \beta_0 + \beta_1 \log(x) + u$$

$$\text{Model C}: \quad y = \beta_0 + \beta_1 x + \beta_2 x^2 + u$$

Consider the following statements:
I. Model A is nested within Model C.

II. Model A is nested within Model B.

III. We can use F test to choose between Model B and Model C.

IV. We can use Adjusted R^2 to compare these models.

Which ones of the statements above are <u>correct</u>?

A. I and II

B. II and III

C. I and III

D. I and IV

E. I, II, and IV.

12. **Consider the following estimation results in equation form:**

$$\widehat{\text{sleephours}} = 8.51 + 0.21 \text{ male} - 0.17 \text{ workhours}$$
$$n = 706, \qquad R^2 = 0.12,$$

where sleephours is the average length of sleep per day (in hours), workhours is the average length of work per day (in hours), and male is a dummy variable that equals 1 if the individual is male and 0 otherwise.

Suppose that we changed the units from hours to minutes by multiplying sleephours and workhours by 60. The new variables are $sleepmins = sleephours \times 60$ **and** $workmins = workhours \times 60$**. Then, the regression equation will be:**

A.
$$\widehat{\text{sleepmins}} = 510.6 + 12.6 \text{ male} - 10.2 \text{ workmins}$$
$$n = 706, \qquad R^2 = 0.12,$$

B.
$$\widehat{\text{sleepmins}} = 510.6 + 12.6 \text{ male} - 10.2 \text{ workmins}$$
$$n = 706, \qquad R^2 = 0.72,$$

C.
$$\widehat{\text{sleepmins}} = 510.6 + 12.6 \text{ male} - 0.17 \text{ workmins}$$
$$n = 706, \qquad R^2 = 0.12,$$

D.
$$\widehat{\text{sleepmins}} = 510.6 + 12.6 \text{ male} - 0.17 \text{ workmins}$$
$$n = 706, \qquad R^2 = 0.72,$$

E. We do not have enough information.

13. **The adverse effects of smoking are well-documented. The following equation summarizes the estimation results of a simple demand for cigarettes.**

$$\widehat{\text{cigs}} = \underset{(2.24)}{6.42} - \underset{(0.10)}{0.85}\,\text{education} - \underset{(0.80)}{0.42}\log(\text{price}) + \underset{(0.25)}{0.38}\,\text{age} - \underset{(0.001)}{0.005}\,\text{age}^2$$
$$n = 500 \quad R^2 = 0.18$$
(standard errors are in parentheses) (1)

where cigs is the number of cigarettes smoked per day, education is the number of schooling years, log(price) is the natural logarithm of cigarette prices, and age is the age of an individual. According to these results, we can say that people start smoking less after the age of .....?

A. 38

B. 76

C. 19

D. 48

E. 42

14. **Consider the following estimation results:**

$$\widehat{\text{IQ}} = \underset{(2.50)}{114.00} + \underset{(0.228)}{3.021}\,(\text{educ}-16) + \underset{(0.0939)}{0.477}\,(\text{pareduc}-32)$$
$$n = 722 \quad \bar{R}^2 = 0.31 \qquad (2)$$
(standard errors in parentheses)

where IQ is the intelligence quotient test score, educ is the number schooling years, pareduc is the sum of parents' education levels. Find the 95% confidence interval for the average IQ prediction at the values educ = 16 and pareduc = 32. (critical value is 2 for simplicity)

A. $[102, 129]$

B. $[100, 109]$

C. $[109, 119]$

D. $[119, 129]$

E. Cannot be found from the information given.